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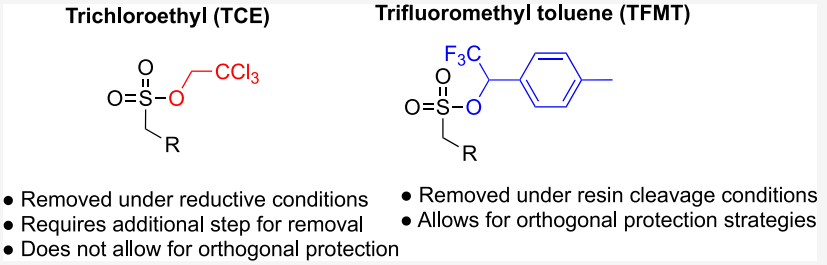
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Synthesis of Sulfonated Peptides Using a Trifluoromethyltoluene-Protected Amino Acid

Ashley E. DeYong, Kaicheng Li, and Elliot L. Chaikof*

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ABSTRACT: A scalable, seven step synthesis is reported for a trifluoromethyl toluene protected sulfonated phenylalanine building block whose utility was demonstrated in the synthesis of four CXCR4-derived sulfonopeptides. When compared to a conventional trichloroethyl protected building block, overall yield was improved by up to 4-fold. We believe this building block will prove to be of significant value for the synthesis of a variety of peptide targets containing phenylalanine sulfonate, a bioisostere of tyrosine sulfate, enabling orthogonal protection strategies and improving synthetic efficiency and yield.

INTRODUCTION

Tyrosine sulfation is a post translational modification that serves as a key regulator of protein−protein interactions, but due to the lability of sulfate groups in acidic or basic environments, sulfopeptides are likely underreported in the proteome.[1] In particular, sulfopeptides are highly susceptible to hydrolysis, which complicates both synthetic and analytical procedures.[2−8] In addition, limited commercial availability of sulfopeptide standards and reagents remains an unmet need for the analysis of this structurally heterogeneous class of compounds. Given the biological significance of sulfopeptides in protein−protein interactions that regulate a variety of biological processes, including cell signaling and immune responses, versatile synthetic schemes for the scalable generation of target compounds are required to enable access to enantiopure materials to explore structure−function relationships and generate functional mimics of potential therapeutic significance.

The acid lability of the ArO−SO_3^- bond presents added complexity in synthesizing sulfated peptides.[9] The sulfate group in combination with the need for acidic conditions to remove both side chain protecting groups and the peptide from the resin can make peptide synthesis difficult, leading to low yields.[10] Additionally, Boc solid phase peptide synthesis (SPPS) is unfit for peptides containing sulfated tyrosine. To address challenges associated with sulfated tyrosine, bioisosteres, such as sulfonates, have been explored as an alternative.[10−13] Sulfonates lack the phenolic oxygen of tyrosine, which is associated with acid-mediated hydrolysis, and various reports describe the preparation of such building blocks for both peptide synthesis and for other sulfate containing classes of compounds, such as steroids and carbohydrates.[14−16] Significantly, in the design of peptide drugs, replacing tyrosine sulfate with a sulfonate group may lead to enhanced biostability and an improved pharmacokinetic profile by limiting hydrolysis and enzymatic cleavage, particularly by sulfatases.[10,11,13] Additionally, sulfonate groups allow for fine-tuning of the drug's pharmacokinetic properties through the choice of the alkyl or aryl substituents attached to the sulfonate, enabling better control over solubility, distribution, and excretion to optimize therapeutic outcomes. Likewise, by carefully selecting the substituents attached to the sulfonate, bioavailability can be increased by reducing polarity and charge, with enhanced membrane permeability and absorption.[17−20]

A number of protecting groups are available for sulfated tyrosine, including neopentyl,[21,22] trichloroethyl (TCE),[23−25] trifluoroethyl,[26] or dichlorovinyl protecting groups.[27,28] In the context of sulfonated phenylalanine, TCE protection has served as a commonly employed strategy. During the course of Fmoc SPPS, TCE can display a tendency to undergo

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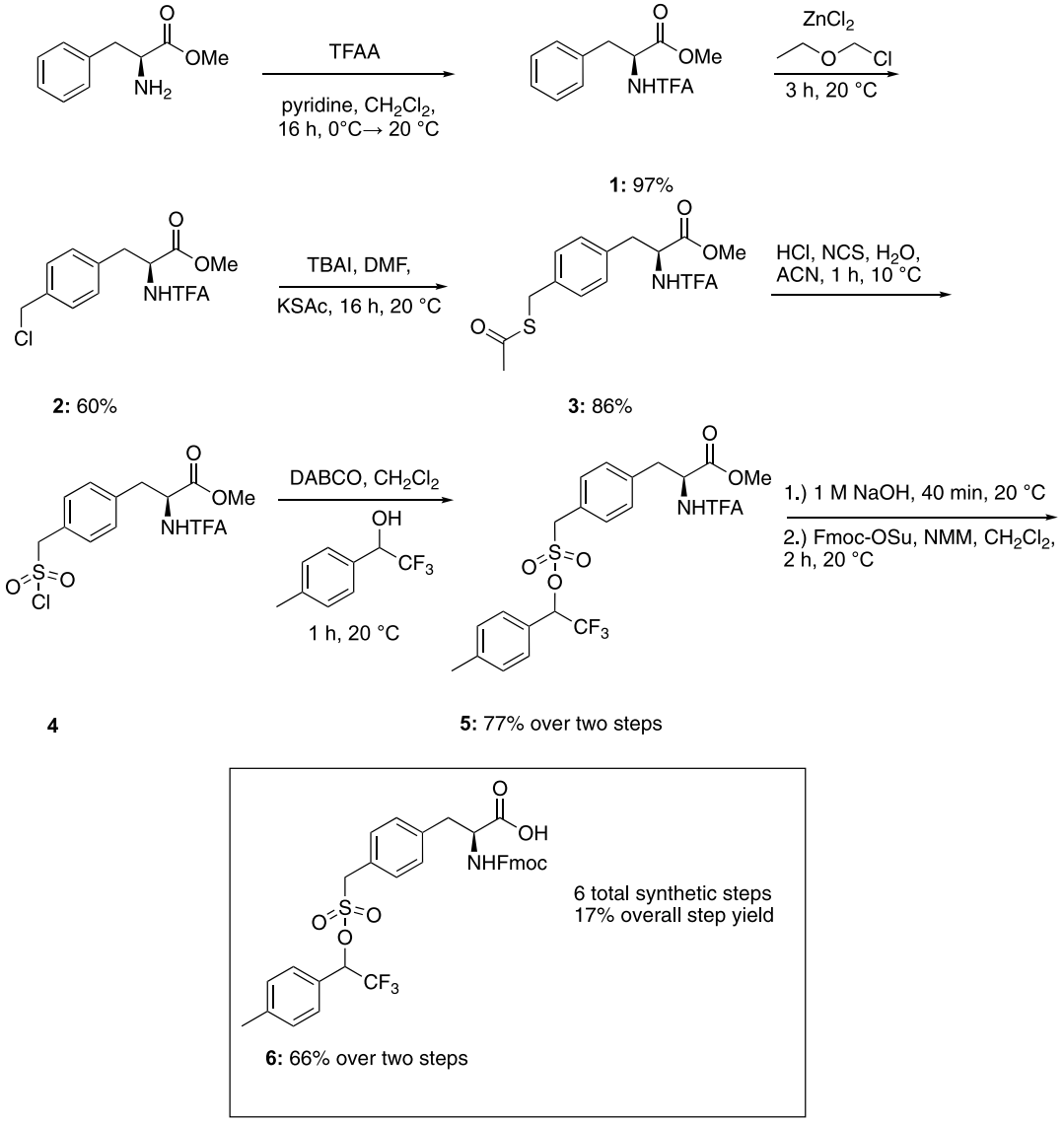


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Scheme 1. Synthetic Pathway to Access TFMT Protected Sulfonated Phenylalanine from L-Phenylalanine Methylester



elimination in the presence of piperidine when used as a protecting group for sulfates.[27] However, elimination does not occur when used as a protecting group for sulfonates.[24] TCE can be removed with mild reductive or acidic conditions using Zn dust in acetic acid, which does not interfere with the stability of Boc or Fmoc groups. Nonetheless, use of TCE is often compromised by inefficient deprotection and potential compatibility issues with other sensitive functionalities. Specifically, incomplete deprotection can lead to purification challenges and significantly impact the yield of the desired product. Removal conditions for TCE, although mild, may also interfere with certain functionalities within the peptide, including groups that are sensitive to reductive conditions.

Identifying a sulfonate protecting group that enables orthogonal protection and can be removed under resin cleavage conditions would provide greater flexibility, improve synthetic efficiency, and potentially limit the number of deprotection steps. In particular, an ideal protecting group would facilitate transformations involving reductive conditions without affecting the protecting group itself. We postulated that trifluoromethyl toluene (TFMT) represented a protecting group that could achieve these goals. While previously described in the synthesis of sulfonate containing oxazine compounds with potential applications in the production of dyes, pharmaceuticals, and specialty chemicals, TFMT has yet to be applied in the context of peptide synthesis.[29]

RESULTS AND DISCUSSION

We sought to develop a synthesis of a TFMT protected sulfonated phenylalanine (Scheme 1). To access thioacetate **3**, we adopted a reported protocol for the synthesis of a TCE protected sulfonated phenylalanine using TFA and methyl groups to protect the amino and carboxylic acid functionalities, respectively.[24] The synthesis started with commercially available, L-phenylalanine methylester, which was first protected with TFA to afford compound **1** in 97% yield. The chloromethyl was then installed via electrophilic aromatic substitution to afford compound **2** in 60% yield. The first three steps proceeded as expected, though further optimization to afford **2** was necessary to make the synthesis more easily scalable.[30] We observed the formation of an undesired disubstituted byproduct with ortho- and para-substitution on

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the phenyl ring, which was mitigated by conducting the reaction for 3 h at 20 °C. The product was isolated by recrystallization rather than flash column chromatography to further support scalable synthesis. Following isolation of compound **2**, a thioacetate was installed via an S_N2 reaction to give compound **3** in an 86% yield.

Compound **4**, the sulfonyl chloride, was then generated from the thioacetate and proceeded immediately to the following coupling step. The TFMT protecting group was synthesized using Ruppert−Prakash conditions,[29,31] and the protecting group was then coupled to sulfonyl chloride **4** to give compound **5** in 77% yield over two steps. As a consequence of the electrophilicity of the sulfonyl chloride and its susceptibility to hydrolysis, anhydrous conditions were required. The TFA and methyl ester protecting groups were removed under 1 M NaOH and upon quenching, the Fmoc protection step was performed to afford **6** in 66% yield over two steps. Biphasic conditions for this last step led to higher overall yields without the need to isolate intermediates. The stoichiometric addition of Fmoc-OSu ensured that an undesired byproduct that could have resulted from hydrolysis of the Fmoc-OSu was avoided.

CXCR4 is a receptor for CXCL12, and dysregulation of the CXCR4/CXCL12 axis and the resultant cellular processes is associated with several disease states, including autoimmunity, immunodeficiency, developmental defects, and over 23 different types of cancers.[32] Previous studies have determined that sTyr21 in CXCR4 enhances CXCL12 dimerization, which suggests that variations in the sulfation pattern of CXCR4 could have functional consequences in vivo.[33] This has led to a variety of efforts to synthesize CXCR4 derived sulfopeptides, as well as other chemokine receptor and ligand derived sulfopeptides to further evaluate structure−function relationships with potential implications for drug discovery.[34−36]

To assess the versatility of the TFMT protecting group, a series of CXCR4 sulfonopeptides were produced (Figure 1). Coupling of the TFMT-protected phenylalanine sulfonate proceeded without difficulty. We observed that removal of the protecting group during resin cleavage needed to be optimized due to the protecting group's tendency to migrate. Various cleavage cocktails were explored, including reagents B and K. Reagent B consists of trifluoroacetic acid, phenol, water, and triethylsilane, whereas reagent K consists of trifluoroacetic acid, water, phenol, thioanisole, and 1,2-ethanedithiol. The migration that was observed involves the TFMT protecting group being cleaved, resulting in a cation, which can then migrate to neighboring aspartic or glutamic acid residues. The resin cleavage cocktails explored contain cation scavengers that limits the protecting group's tendency to migrate. Migration was observed with reagent B but minimal migration noted with reagent K. Cleavage cocktails were assessed using HPLC. With reagent B, 93% purity was noted, whereas with reagent K, 97% purity was noted. This can likely be explained by reagent K containing a more nucleophilic scavenger, thereby, enhancing the efficiency of reducing the migration reaction.

Four sulfonopeptides **7**, **8**, **9**, and **10** were synthesized using Fmoc SPPS on Rink Amide resin in 38−56% yield. Given the stability of TFMT under reductive conditions, we postulated that it would be possible to selectively remove Alloc, a commonly used amino protecting group, to confirm the capacity of TFMT to serve as an orthogonal protection strategy for sulfonate groups (Scheme 2). Peptide **11**, as a biotinylated analogue of **10**, was synthesized in 50% yield, confirming the effectiveness of this strategy.

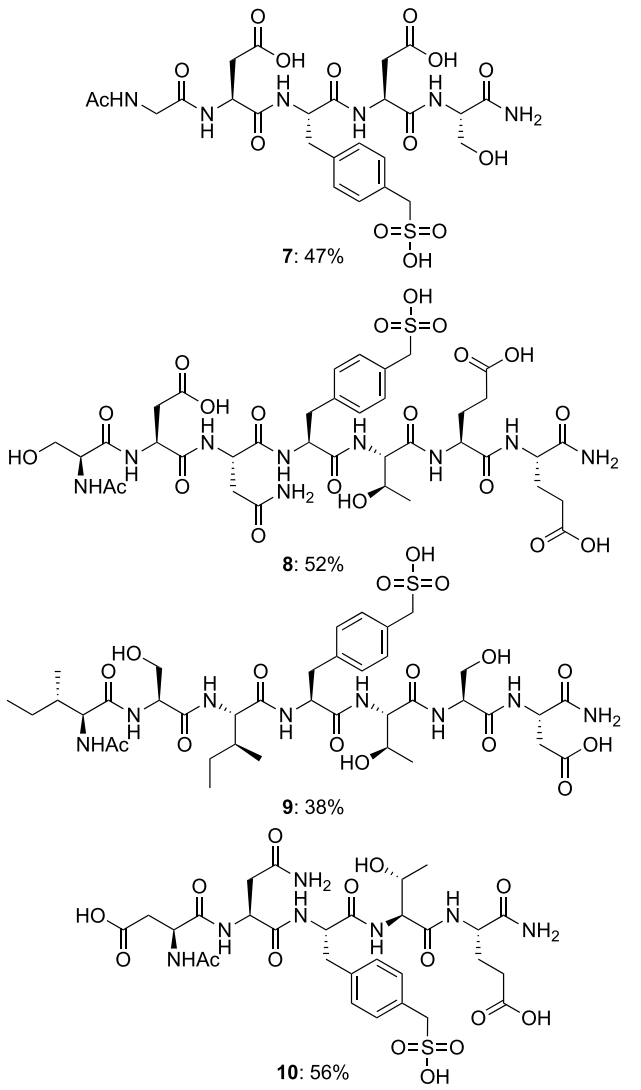


Figure 1. Synthesis of a variety of sulfonopeptides using a TFMT protected building block.

In addition to facilitating orthogonal protection, improvement in overall yield was also observed with the use of TFMT as compared to the TCE protecting group. When TCE was employed to produce **10** rather than TFMT, hydrogenation was required to deprotect the peptide with an overall yield of 13% over two steps (resin cleavage and hydrogenation), a 4-fold lower yield than that achieved with TFMT (Scheme 3).

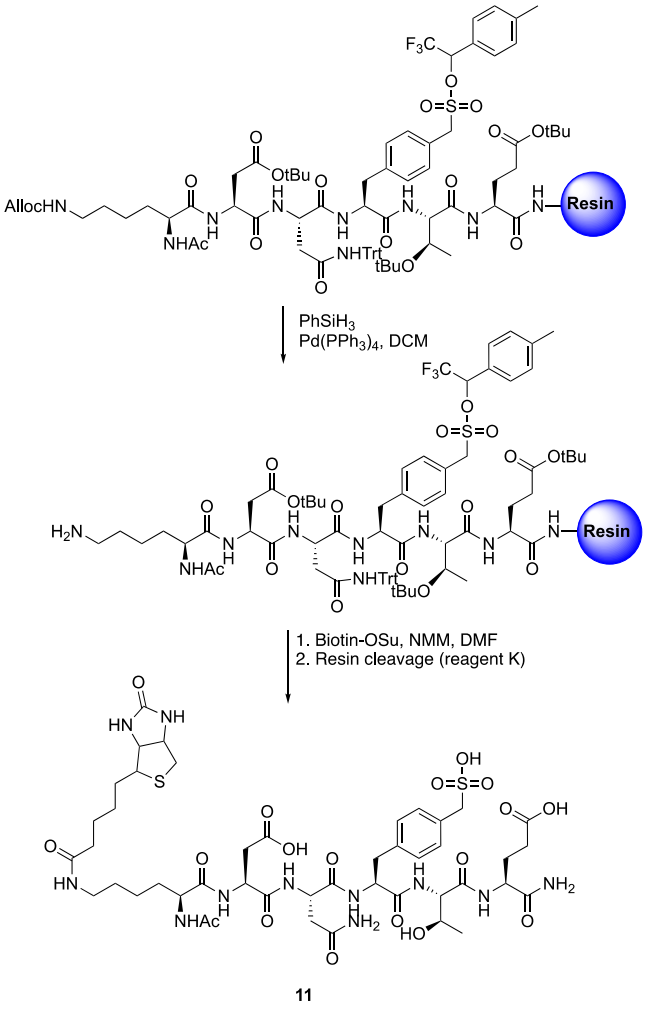
CONCLUSION

In conclusion, TFMT can be readily incorporated as a means to protect sulfonated amino acids for facile synthesis of sulfonopeptides. TFMT can be removed under standard resin cleavage conditions with TFA and is resistant to reductive conditions, thereby facilitating orthogonal protection of additional functional groups. We demonstrate that the use of TFMT can shorten synthesis and significantly improve yield, while enabling orthogonal protection as illustrated by on-resin installation of biotin, as one example. While the utility of TFMT as a protecting group was demonstrated in the synthesis of sulfonopeptides, we believe its application will also prove to be advantageous for generating other biologically

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Scheme 2. Demonstration of an Orthogonal Protection Strategy that is Enabled by a TFMT-Protected Amino Acid



important classes of compounds, such as sulfated glycans and steroids.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02529.

Full experimental details; ^{1}H NMR, ^{13}C NMR, and HRMS of all new compounds (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Elliot L. Chaikof − *Department of Surgery, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, Massachusetts 02215, United States; Division of Health Sciences and Technology, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, United States; Broad Institute of MIT and Harvard, Cambridge, Massachusetts 02142, United States;* orcid.org/0000-0001-6184-4571; Phone: (617) 632-9581; Email: echaikof@bidmc.harvard.edu; Fax: (617) 632-9701

Authors

Ashley E. DeYong − *Department of Surgery, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, Massachusetts 02215, United States*

Kaicheng Li − *Department of Surgery, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, Massachusetts 02215, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.4c02529

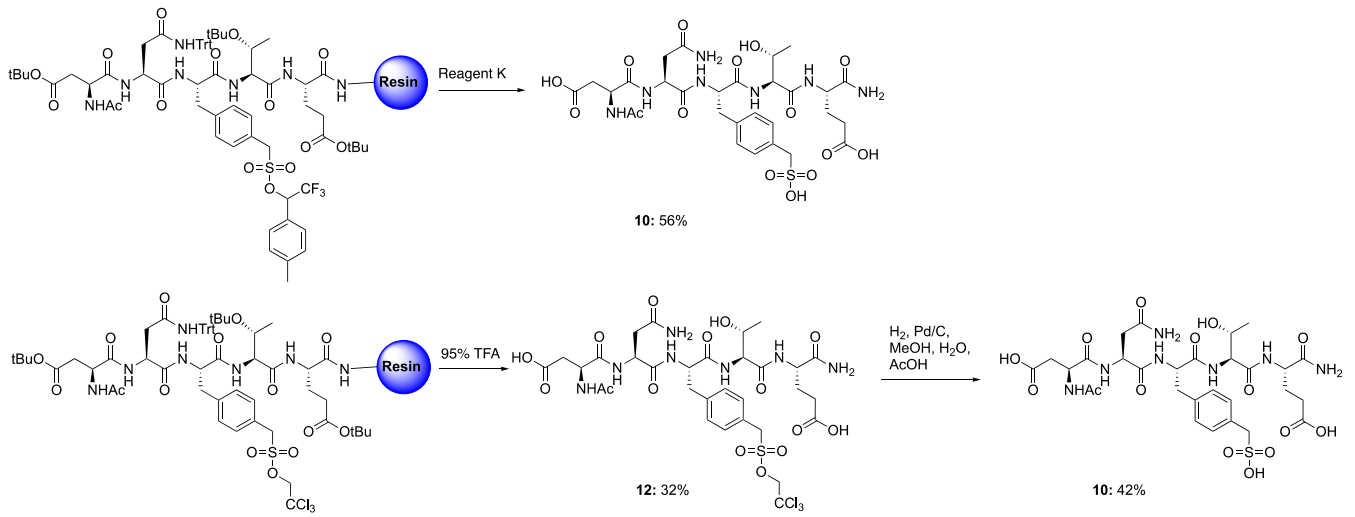
Author Contributions

A.E.D. contributed to design of synthetic route and synthesized and characterized all compounds in this work. K.L. contributed to design of synthetic route. E.L.C. contributed to the design of experiments and analysis of results. A.E.D., K.L., and E.L.C. wrote the manuscript.

Notes

The authors declare no competing financial interest.

Scheme 3. Hydrogenation of Compound 12 to Give Fully Deprotected Sulfonopeptides and Comparison of TFMT and TCE Protecting Groups



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